ai.py - C:\Users\ganes\AppData\Local\Programs\Python\Python313\ai.py (3.13.3)

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```
from collections import deque

def solve(start, goal):
    queue = deque([(start, [])])
    visited = {start}

    while queue:
        state, path = queue.popleft()

        if state == goal:
            return path + [state]

        zero = state.index(0)
        row, col = divmod(zero, 3)

        for dr, dc in [(0, 1), (0, -1), (1, 0), (-1, 0)]:
            r, c = row + dr, col + dc

            if 0 <= r < 3 and 0 <= c < 3:
                new_pos = r * 3 + c
                new_state = list(state)
                new_state[zero], new_state[new_pos] = (
                    new_state[new_pos], new_state[zero]
                )
                new_state = tuple(new_state)

                if new_state not in visited:
                    visited.add(new_state)
                    queue.append((new_state, path + [state]))

    return None

start = (1, 2, 3, 4, 0, 6, 7, 5, 8)
goal = (1, 2, 3, 4, 5, 6, 7, 8, 0)

solution = solve(start, goal)

if solution:
    for state in solution:
        print(state[:3])
        print(state[3:6])
```

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IDLE Shell 3.13.3

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```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\ganes\AppData\Local\Programs\Python\Python313\ai.py ====
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)

>>>
```